FINFLOW

Buy or rent a 2-bedroom apartment in Hanoi?

A tool that checks your financial safety and compares net worth when buying vs renting

GROUP G04

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Course: NHA408E · **Lecturer:** Phan Tran Trung Dung

THE JOURNEY IN 5 STEPS

1 Your finances
How much can we afford?

2 Project & size
Where do we want to live?

3 Choose units
Which unit exactly?

4 Invest spare money
Where does the surplus go?

5 RESULT
Buy or rent, and how safe?

756
for-sale listings

277
for-rent listings

3
districts: Cau Giay, Ha Dong, Tay Ho

2-bedroom units · batdongsan.com.vn, 9 Sep 2026 · one HTML file, runs in the browser. Limits: asking prices, not sale prices; data does not update itself.

Our demo household

- Couple aged 32 and 30
- Income 35M + 25M VND/month
- Savings 2.5B VND
- Living cost 15M VND/month
- Emergency fund: 6 months

1 YOUR FINANCES

- Inputs: incomes, savings, living cost, existing debt, emergency fund, ages, promo package (MB Bank 6/12/18/24 months fixed)
- **Loan term =** min((70 - older age) × 12, 360 months)
- Inputs checked first: living cost < income; loan term > 24-month grace period
- **"Your budget" appears as you type** — same engine as the final result

Buy ≈ 4.3B · Rent ≈ 26.2M/month
166/756 for sale and 273/277 rentals within budget

YOUR BUDGET

Buy up to about **4.3B VND** (typical 72 m² unit). Limit: your **monthly payments** — above this price, the tightest month leaves less than your safe level of income.

Rent up to about **26.2M VND/month** (typical 80 m² unit), plus the service fee.

166 of 756 homes for sale and **273** of 277 rentals are within budget (536 payments too tight · 54 short of savings · 4 rents too tight).

▼ **Where are the units within budget? (district × size)**

Each cell: within budget / all units, and the median price (buy in billion VND, rent in million VND per month).

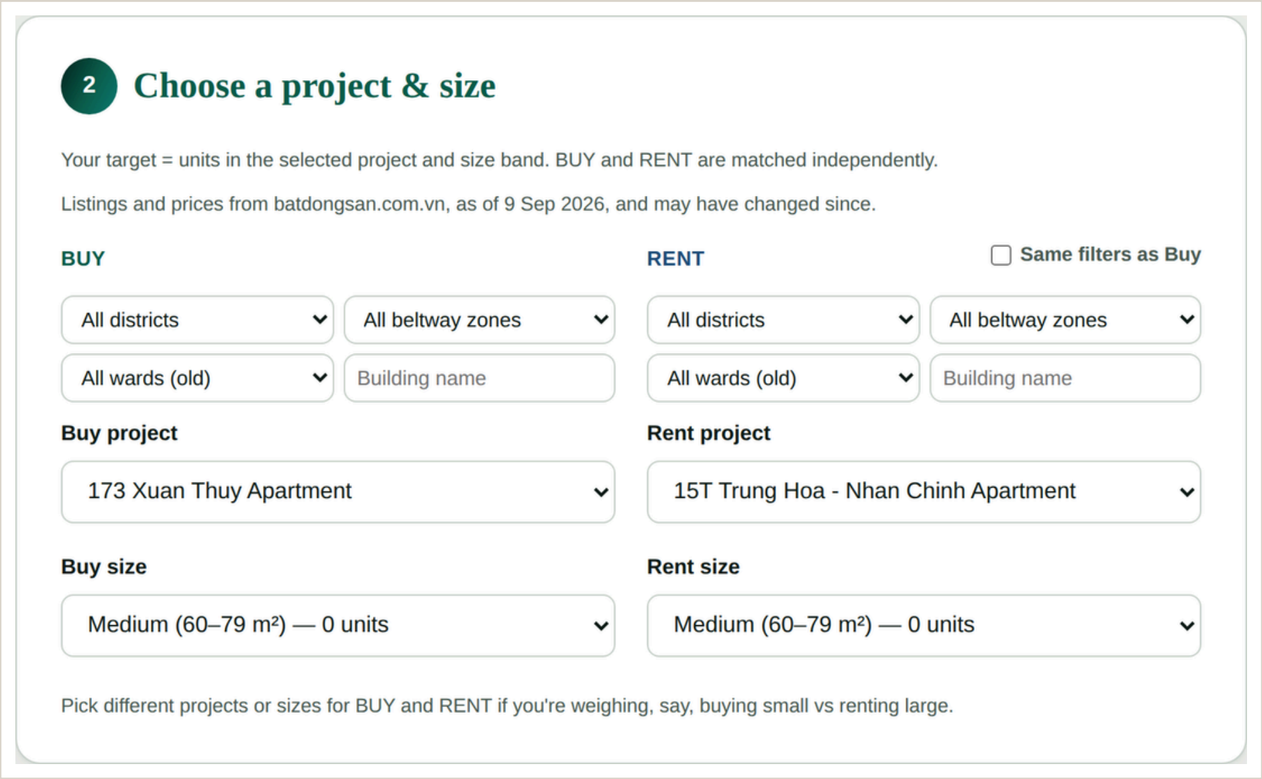
Buy	Small	Medium	Large
Cau Giay	**23**/47 4.5B	**10**/155 6.8B	**0**/107 8.3B
Ha Dong	**61**/63 3.3B	**66**/176 4.6B	**6**/45 5.5B
Tay Ho	**0**/9 6.3B	**0**/62 7.7B	**0**/92 10.6B

Rent	Small	Medium	Large
Cau Giay	**7**/7 14.0M	**48**/48 16.0M	**73**/74 17.0M
Ha Dong	**3**/3 9.0M	**42**/42 11.0M	**17**/17 12.5M
Tay Ho	**3**/3 12.0M	**18**/18 14.0M	**62**/65 19.0M

Same check as the final result (Stable scenario, 12-month promo package, 6-month emergency fund kept aside), including the service fee and, for owners, fire insurance. Prices and rents are for a typical size; each unit's label in Step 3 is checked for that exact unit.

2 CHOOSE A PROJECT & SIZE

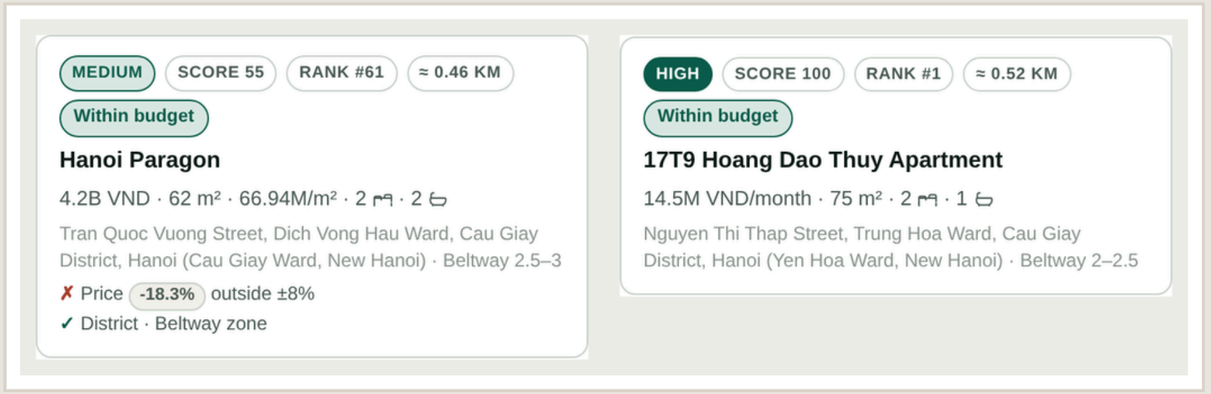
- Filter by district, beltway zone, old ward or building name
- **Project + size band = your TARGET**
- Small 45–59 m² · Medium 60–79 m² · Large 80–100 m²
- BUY and RENT are chosen independently — e.g. buy small here vs rent larger elsewhere
- “Same filters as Buy” when you want both sides alike



3 CHOOSE A BUY UNIT AND A RENT UNIT

42% of BUY project × size combos have just 1 listing → the app suggests similar units

- **Price anchor = median:** target if ≥ 3 units, else whole size band
- **Tolerance:** Small ±5% · Medium ±8% · Large ±10% (in-building variation ≈ 6.4%)
- Ties → closest to the chosen project first
- Each card shows ✓/✗ per criterion; “Only within budget” hides the rest



Price 45	District 30	Beltway 25

HIGH 100	MEDIUM 55–75	LOW 25–45	WARNING 0

- **Within budget**
- **Payment too tight**
- **Not enough savings**
- **No loan available**

4 HOW YOU INVEST SPARE MONEY

Savings deposits

6.50% /year avg

Government bonds

4.02% /year avg

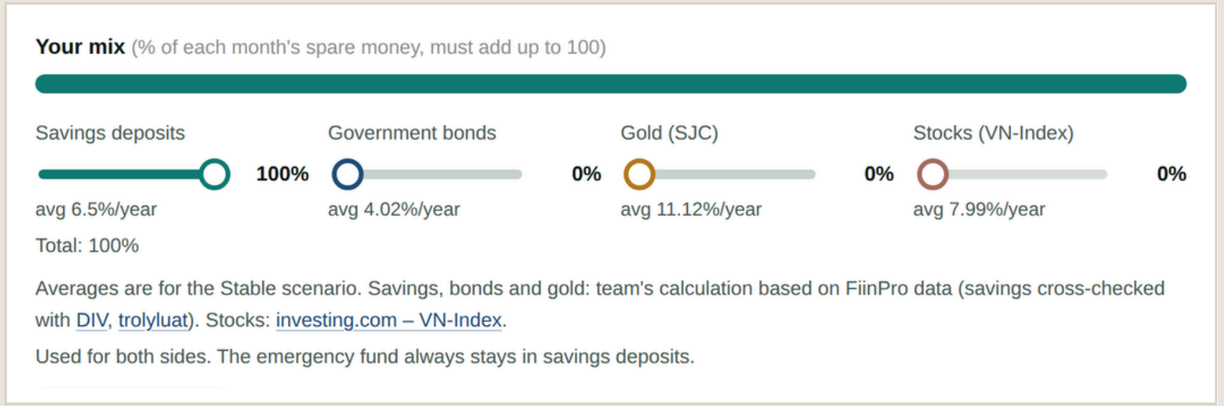
Gold (SJC)

11.12% /year avg

Stocks (VN-Index)

7.99% /year avg

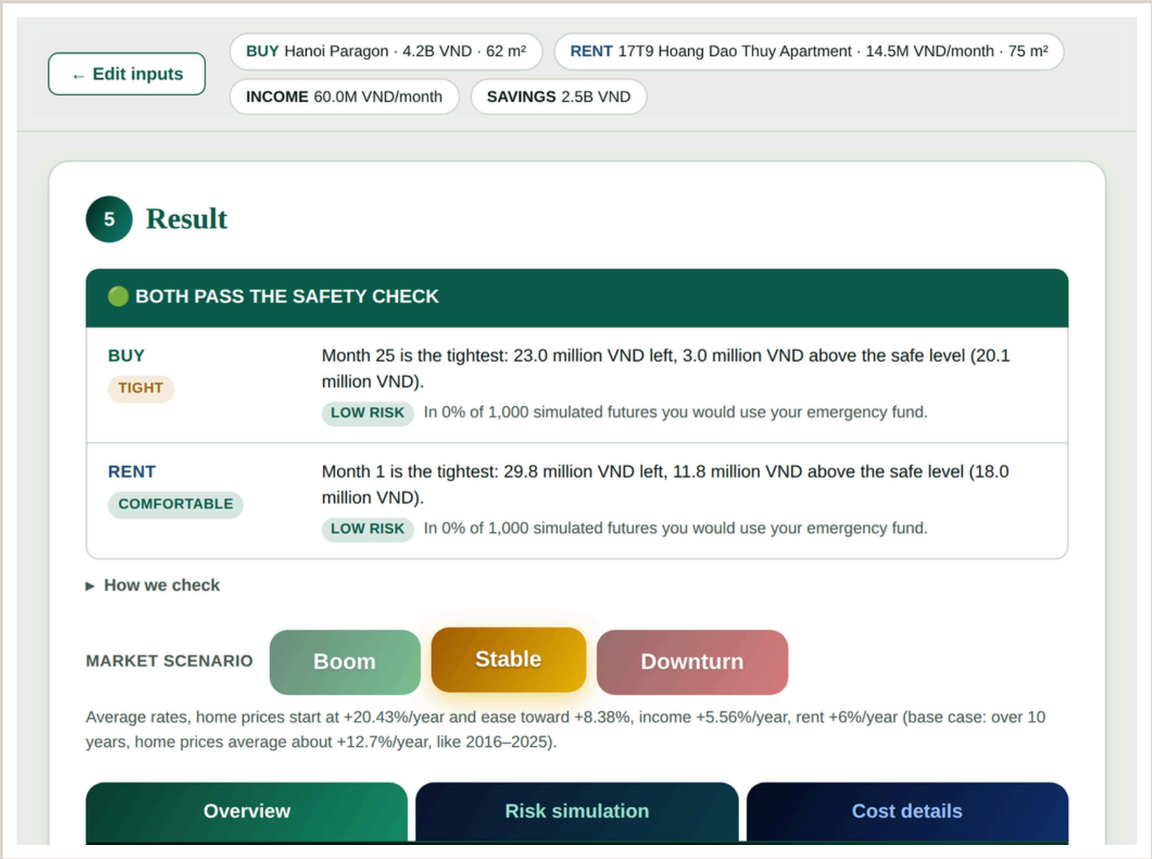
- **Default: 100% savings** — neutral, no risk appetite assumed
- Mix must add up to 100%; rebalanced every year
- Same mix for BUY and RENT → a fair comparison
- Emergency fund always stays in savings, never used for the down payment
- Sources: FiinPro (savings, bonds, gold) · investing.com (VN-Index)



5 THE VERDICT & THE SAFETY CHECK

- **Opens as its own screen** — "← Edit inputs" goes back with everything kept
- Headline verdict, then each side rated Comfortable / Tight / Not safe
- **Checks all 360 months** and reports the tightest one
- **Safe level:** money left ≥ 40 / 35 / 30 / 25% of income, by income tier
- Market scenario: Boom · Stable · Downturn — everything recalculates
- Details in 3 tabs: Overview · Risk simulation · Cost details

Our couple: both pass
BUY Tight — month 25: 23.0M left vs safe level 20.1M
RENT Comfortable — 11.8M above the safe level



Result tab: **Cost details**

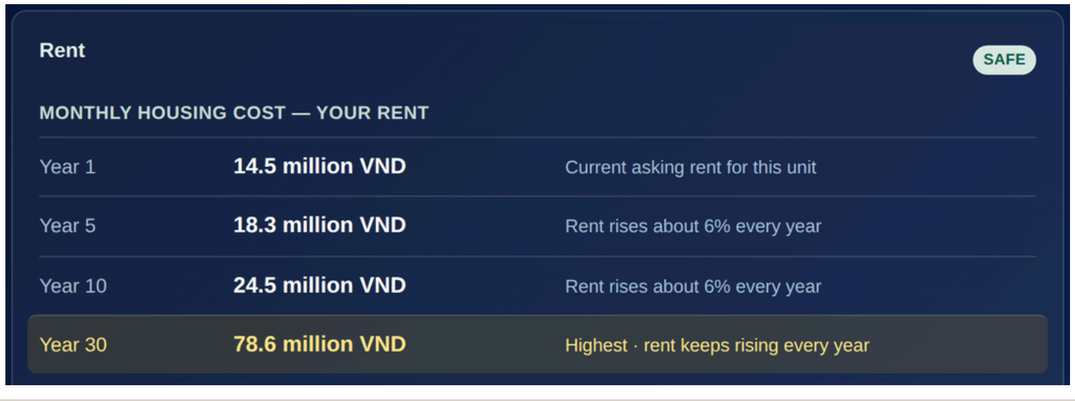
5 WHERE THE MONEY GOES EACH MONTH

BUY — bank payment by phase



12.8M (months 1–12, fixed 8.3%) → 18.5M (13–24, floating, interest only) → 27.1M from month 25

RENT — rises every year



- **Month 25:** grace ends, principal starts → 27.1M, then falls to 5.6M
- Plus running costs: service fee + fire insurance ≈ 0.8M/month
- Emergency fund needed? Run out of money? — never, for our couple

5 REALITY CHECK & NET WORTH OVER TIME

Reality check

Reality check: rent vs price

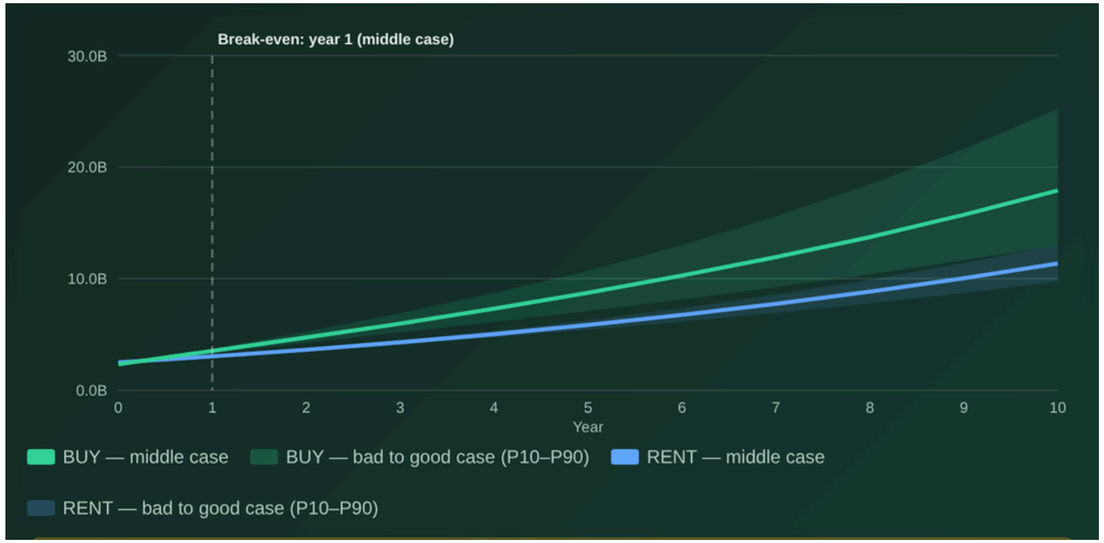
Gross rental yield (per m², this pair)	Price-to-rent (years of rent to pay the price)	Yield in year 10 under these assumptions
3.5%	**29 years**	**1.8%**

Each year, rent pays **3.5%** of the price per m² (4.2% as the two units are listed). That's low: the market median is **2.3%** in our data and **3.7%** for 2-bedroom units in Hanoi (Global Property Guide, Q3/2025), while savings earn **6.5%** and the loan costs **12%**, then **14%** after the promo. Buying comes out ahead only if home prices keep rising (**20.43%** easing to **8.38%**, vs rent **+6%**). Check the Downturn scenario and the sensitivity table.

Rent pays only 3.5% of the price a year (29 years of rent) → buying wins only if prices keep rising

BUY = home value - 1.5% selling cost - loan - prepayment fee + investments
RENT = savings + monthly surplus, invested

Net worth over time



- Middle case + P10–P90 band · break-even marker
- Compare at 10 / 20 / 30 years · nominal or today's money

Result tab: **Risk simulation**

5 1,000 SIMULATED FUTURES

97% of futures: BUY richer than RENT after 10 years (our couple)

Emergency fund needed: BUY 0% · RENT 0%
Net worth P10 / P50 / P90
BUY 13.0 / 17.9 / 25.2B · RENT 9.7 / 11.4 / 13.1B

- Rates, home prices, rent, income and returns vary at random (stocks σ 19.30, gold σ 14.13)
- Run in the page · fixed seed · under 1 second
- Sensitivity table: which assumptions could flip the answer



Not just "buy" or "rent" — how safe each choice is, and how sure we can be.

Thank you — questions?